

Cover Letter for IEEE Submission

Date: September 18, 2026

To:

Editor-in-Chief

IEEE Transactions on Engineering Management (IEEE TEMS) /

IEEE Transactions on Systems, Man, and Cybernetics: Systems (IEEE SMC)

Subject: Submission of Original Research Manuscript for Double-Blind Peer Review

Dear Editor-in-Chief,

I am pleased to submit our original research manuscript titled “**Value Chain Physics: Formal Proof and Industrial Validation Based on Non-IID and Qian Xuesen’s Open Complex Giant Systems**” for consideration as a Regular Paper in *IEEE Transactions on Engineering Management* (or *IEEE Transactions on Systems, Man, and Cybernetics: Systems*).

Background and Core Scientific Contribution

Traditional industrial engineering and supply chain planning systems (such as MRP and APS) rely heavily on Independent and Identically Distributed (IID) and linear decoupling assumptions. In real-world enterprise networks characterized by Non-IID topological strong coupling, these traditional models inevitably suffer from computational breakdown and cascading failures.

This paper bridges complex systems science and industrial operations engineering by formalizing **Qian Xuesen’s Open Complex Giant Systems (OCGS)** theory:

1. **First Principle Formulation:** We establish **Non-Identical and Non-Independently Distributed (Non-IID)** coupling as the first principle of Value Chain Physics (VCP), deriving a qualitative Eight-Axiom System spanning Teleology to Evolution on a five-dimensional orthogonal topological manifold $\mathcal{M}(t)$.
2. **Formal Proof via Banach Fixed-Point Theorem:** We prove **Theorem 1 (Closed-Loop Convergence Theorem for Non-IID Complex Giant Systems)**. By introducing a DAG topological pruning operator Π_{cut} operating in $O(N\log N + K)$ time, the non-convex domain is mapped into a locally strongly convex closed subset $\Omega_{\text{convex}}^{\Delta t}$. In a complete normed Banach space, we prove the existence and uniqueness of a self-consistent **structural fixed point S^*** with explicit contraction constant $k = 1 - \frac{\gamma\Delta t}{\|Q\| + \|R\|} \in (0,1)$, completing the mathematical proof of computability for Qian Xuesen’s OCGS.
3. **18-Month Large-Scale Industrial Validation:** We validate the theory on Lenovo’s Global Integrated Planning System (IPS) and LCFC (WEF Lighthouse Factory), daily solving 50,000 backlog orders, 2,000,000 SKU-Site nodes, and 150,000 physical constraints. The results confirm a silicon-based autonomous decision-making ratio of **over 95.0% for rigid closed-loop automation**, with the 48-

hour delivery commitment response rate surging from 50%–60% to **over 98.0%**, inventory dropping by 50%, and turnover increasing by 1.9x.

Double-Blind Peer Review and Ethical Declarations

In full compliance with IEEE double-blind peer review policies: * The submitted manuscript (Paper_1_Formal_Proof_OCGS_Qian_Xuesen_EN_DoubleBlind.pdf) has been completely anonymized, removing all author names, affiliations, headers, footers, and biographical references. * Author metadata and declarations are provided separately in the Title Page document (IEEE_Submission_Title_Page.pdf). * This manuscript represents original work, has not been published previously, and is not currently under consideration by any other journal or conference. * The author declares no financial or personal conflicts of interest.

We believe this paper provides a groundbreaking theoretical foundation and empirical validation for intelligent industrial planning systems, which will be of strong interest to the readers of IEEE TEMS and IEEE SMC.

Thank you very much for your time, consideration, and management of the peer-review process.

Sincerely,

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